

PARAALLAX OFFICIAL DOCUMENTATION

PENALTY SYSTEM GUIDE

COMPLETE REFERENCE FOR ALL GAME TYPES

VERSION

V1.0

GAMES

7 MODULES

AUDIENCE

PLAYERS

STATUS

ACTIVE

⚠ PLAYER MANUAL - PENALTY RULES ARE ENFORCED IN REAL TIME ⚠



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INTRODUCTION & OVERVIEW

The **Paraallax Penalty System** is an integral part of the game experience. Every mistake you make across all 7 puzzle modules adds time to your team's total penalty counter, which is deducted from your final score when the session ends.

Penalties are applied **instantly and in real time** — the moment a wrong action is taken, the server records the penalty. There are **no warnings** before the penalty fires, and they **cannot be undone**. Understanding exactly what triggers a penalty in each game is therefore critical to achieving a top score.

This manual documents every penalty trigger, its mechanism, and the exact amount added to your team's penalty counter for each of the 7 game types in Paraallax.



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HOW PENALTIES WORK

PENALTY ACCUMULATION

- ▶ Penalties are tracked per **team**
- ▶ They accumulate in **seconds**
- ▶ Each trigger adds a fixed **penalty block**
- ▶ Default block = **5 minutes (300s)**
- ▶ Admin can configure penalty time
- ▶ Penalties stack with no upper limit

PENALTY CALCULATION

- ▶ The **Session** holds `penaltyMinutes`
- ▶ Default session penalty = **5 min / event**
- ▶ API endpoint: **POST /api/team/add-penalty**
- ▶ Penalty is added to **team.penaltySeconds**
- ▶ Displayed on admin dashboard in real time
- ▶ Included in the final results page

$$\text{Final Score} = \text{Puzzles Solved} - \text{Total Penalty Seconds}$$

Lower penalty seconds = better rank. Each wrong action adds penalty time to your total.



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QUICK-REFERENCE PENALTY SUMMARY

GAME MODULE	SETS	PENALTY TRIGGER	PENALTY PER EVENT
GAME 1 – Power Grid	3	All 4 gates placed but circuit is wrong	1× penalty block
GAME 2 – Handshake Protocol	2	Progress drops to 0 for every 2nd failure	1× penalty block
GAME 3 – Database Schema	3	Wrong/incomplete SQL query submitted	1× penalty block
GAME 4 – The Firewall	3	Wrong move (-15s local) OR time runs out	1× penalty block (timeout only)
GAME 5 – OS Deadlock	3	Wrong kill command OR invalid command	1× penalty block
GAME 6 – Dynamic Cipher Grid	3	Wrong cipher answer submitted	1× penalty block

GAME MODULE	SETS	PENALTY TRIGGER	PENALTY PER EVENT
GAME 1 – Memory Access System	3	Wrong request output OR wrong final answer	1× penalty block

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GAME 1 – THE POWER GRID

GAME 01

The Power Grid

3 SETS · Grid Alpha / Beta / Gamma

- Trigger:** All four gate slots (Z1, Z2, Z3, Z4) have been filled with logic gates, *but the final output node does not produce a `TRUE` signal*. The circuit is evaluated live and the penalty fires automatically the moment a complete-but-wrong configuration is detected.

+1 PENALTY
BLOCK ADDED
- One-shot protection:** A state flag (`hasTriggeredPenalty`) ensures the penalty fires only *once per complete wrong configuration*. If you remove a gate and re-place it, the flag resets and another wrong configuration will trigger another penalty.

⚠️ RESETS ON
GATE REMOVAL

MECHANIC: Four draggable gates (Δ AND · Σ OR · Φ XOR · Ψ NAND) must be arranged so the full circuit produces TRUE at the output node. Each gate combination is checked instantly. Removing a gate from any slot resets the error-guard, allowing repeated experimenting – but each wrong complete arrangement costs one penalty block.

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GAME 2 – THE HANDSHAKE PROTOCOL

GAME 02

The Handshake Protocol

2 SETS · Protocol Alpha / Beta

- Wrong button press:** Pressing the wrong action button (`ACTION A` , `ACTION S` , or `IGNORE`) for the current hex code decreases your progress bar by **-2 points** and marks the current code as `[ERR]`.

-2 PROGRESS
NO PENALTY YET
- Missed code (timeout):** Each code has a **2.5-second window**. If the code expires without any action, it counts as a failure and progress drops by -2, same as a wrong press.

-2 PROGRESS
NO PENALTY YET
- Penalty block trigger — the "wipe" rule:** A penalty block is only sent to the server when the progress counter is driven *from above 0 all the way back to 0*, AND this happens for the **2nd, 4th, 6th ... (every even-numbered) wipe**. In other words: the internal `penaltyCountRef` increments on each wipe, and only fires a server penalty when that count is even (`penaltyCount % 2 == 0`).

+1 PENALTY
EVERY 2ND WIPE

MECHANIC: Codes scroll every 2.5 s. Reach 10 correct signatures to win. Protocol Alpha (Set 1): ACTION A if `hex % 7 == 0` · ACTION S if hex contains 'C' or 'D' · else IGNORE. Protocol Beta (Set 2): ACTION A if `(digit1 + digit2) > 15` · ACTION S if both hex digits are equal · else IGNORE. Keys: [A] = Action A · [S] = Action S · [I] = Ignore.

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GAME 3 – DATABASE SCHEMA

GAME 03

Database Schema

3 SETS · Emergency Override Console

- Missing SELECT or FROM keyword:** If your typed query does not include both the word `select` and `from`, the system immediately rejects it with `ERROR 1064: SQL syntax error` and a penalty block fires.

+1 PENALTY
BLOCK ADDED
- Missing semicolon:** Any query that includes SELECT and FROM but *does not end with a semicolon* (`;`) is rejected with a syntax error and a penalty block is added.

+1 PENALTY
BLOCK ADDED
- Syntactically valid but wrong query:** A query that is structurally correct (has SELECT, FROM, ends with ';') but *does not match the expected answer* receives the message `ERROR 0000: QUERY EXECUTED BUT RETURNED NO MATCHING INCIDENT DATA` and triggers a penalty block.

+1 PENALTY
BLOCK ADDED

MECHANIC: Each set presents a different incident scenario. You must compose an exact SQL SELECT query matching the expected query (case-insensitive, extra spaces ignored, single/double quotes are equivalent). Read the incident prompt carefully – the table name, column names, and WHERE clause must all be correct. Example Set 1 answer: `select first_name, last_name from personnel where emp_id = 4091;`

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GAME 4 – THE FIREWALL

GAME 04

The Firewall

3 SETS · Sector Alpha / Beta / Gamma · 7×7 Grid

- Wrong move (local penalty — in-game timer only):** Each incorrect directional move causes a `COLLISION DETECTED` event. The local countdown timer is *immediately reduced by 15 seconds* and your position is **reset to the start** after an 0.8-second stun. *No server penalty is added for individual wrong moves.*

-15 SECONDS
LOCAL TIMER ONLY
- Running out of time (timeout):** If the 2-minute countdown (**120 seconds**) reaches zero before you reach the target coordinate, the game ends in `CONNECTION TERMINATED`. At this point, **one full penalty block is added to the server**. You may retry by clicking "RETRY BREACH" — but each timeout costs a penalty block.

+1 PENALTY
ON TIMEOUT ONLY
- Move logic (same for all 3 sets):** At any position (X, Y), the valid move is determined by: if `(X+Y) is even` → move UP if X<Y, else RIGHT. If `(X+Y) is odd` → move LEFT if X<Y, else DOWN. Any deviation triggers a collision and -15s.

RULE-BASED
MOVEMENT

MECHANIC: Grid is 7×7. Controls via arrow keys or WASD or on-screen D-Pad. The collision penalty (-15s) is visual and local only; it serves as your countdown punishment. The server-side penalty block is only sent when the full timer expires (timeout = game failure). Moves are throttled to 150ms intervals to prevent key-spam accidents.

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GAME 5 – OS DEADLOCK

GAME 05

OS Deadlock

3 SETS · PARALLAX OS v5.0 · Process Terminal

- Wrong kill target:** Entering `kill <PID>` with a process ID that is *not the correct* `root blocker` prints `Incorrect process terminated ❌` and `Deadlock persists...`, and adds one penalty block.

+1 PENALTY
BLOCK ADDED
- Invalid command:** Typing anything that is *not a recognised command* (e.g. random text, typos, wrong syntax) prints `Invalid command` and immediately fires a penalty block.

+1 PENALTY
BLOCK ADDED

MECHANIC: On startup the terminal auto-runs `ps` (process list), `resource-map` (resource lock table), and `waiting-list` (blocked processes) commands. Study the output: find which process holds the resource that is blocking the most other processes (highest usage, most dependents). Then type: `kill <PID>`. No other commands are needed – typing unrecognised commands wastes a penalty. Set 1 answer: `kill 344` (CACHE_LOCK at 95% usage, blocking 2 processes).

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GAME 6 – DYNAMIC CIPHER GRID

GAME 06

Dynamic Cipher Grid

3 SETS · Cipher Decryption · Submit Final Word

- Wrong cipher answer submitted:** Clicking the `SUBMIT` button (or pressing `Enter`) with an answer that does not match the expected decoded word triggers `❌ Access Denied` and adds one penalty block. The input field shakes visually as feedback. A new attempt is allowed immediately.

+1 PENALTY
EACH WRONG SUBMIT
- No penalty for empty submission:** Clicking `Submit` with an empty input is blocked by the UI and does *not* trigger any penalty.

SAFE
EMPTY = BLOCKED

MECHANIC: Each set shows a 3x3 grid of code cells (e.g., A1, C3, F2...), a sequence of those codes, a SHIFT value, and a MODE (ALPHA). Follow the cipher steps from your game guide: (1) look up each code in the grid to find its row/column letter, (2) apply the SHIFT value to the alphabet, (3) assemble the decoded letters in sequence order. The answer is always a 5-letter word. Input is case-insensitive (converted to uppercase on submission). Think carefully before submitting.

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GAME 7 – MULTI-REQUEST MEMORY ACCESS SYSTEM

- **Stage 1 — Wrong request output:** Each set has 3 binary memory requests, processed one at a time in order. Submitting the wrong decoded letter for the **PENDING** (active) request fires the message **X INVALID REQUEST OUTPUT** and adds one penalty block. The same request remains active for retry — but each wrong attempt costs a penalty.

 +1 PENALTY
EACH WRONG LETTER

- **Stage 2 — Wrong final assembly:** After all 3 requests are **SOLVED**, you must assemble the 3 output letters in the order specified by the **Final Rule** (e.g. 1st+3rd+2nd). Submitting the wrong 3-letter word fires **X INVALID FINAL ASSEMBLY** and adds one penalty block.

 +1 PENALTY
EACH WRONG FINAL

- **Stage 1 is sequential — locked requests cannot be skipped:** Requests are locked until the previous one is solved. You can only interact with the current **PENDING** request. Wrong answers for it still cost a penalty.

 **SEQUENTIAL
ORDER ENFORCED**

MECHANIC: Stage 1 – each binary string (e.g. 101101) must be decoded to a single letter using the binary-to-letter mapping in your guide manual. Stage 2 – rearrange the 3 decoded letters according to the Final Rule shown on screen (e.g. "1st + 3rd + 2nd"). Set Alpha example: REQ1=101101→R, REQ2=010010→M, REQ3=110100→A. Rule: 1st+3rd+2nd → "RAM". Input is single-character for Stage 1 and 3-character for Stage 2. Both are case-insensitive.



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STRATEGY & TIPS

✓ DO THIS

- ✓ Read all on-screen data before acting
- ✓ In Game 6, solve the cipher offline first
- ✓ In Game 4, memorise the movement rule before starting
- ✓ In Game 2, stay calm — wrong keys still cost progress
- ✓ In Game 3, always end your SQL with a semicolon
- ✓ In Game 7, decode all 3 letters before assembling
- ✓ Aim for zero wrong submissions in Game 6 & 7

✗ AVOID THIS

- ✗ Guessing randomly in Game 6 or 7 — every wrong submit costs
- ✗ Running out of timer in Game 4 — each timeout = full penalty
- ✗ Typos in Game 3 (missing semicolons, wrong table names)
- ✗ Typing unrecognised commands in Game 5
- ✗ Pressing wrong keys repeatedly in Game 2
- ✗ Testing all gate combos blindly in Game 1
- ✗ Not reading the Final Rule in Game 7 before assembling



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GLOSSARY OF TERMS

Penalty Block	A fixed amount of penalty time added to your team's total. Currently configured by the admin per session (default: 5 minutes = 300 seconds). One "block" is added per penalty event.
penaltySeconds	The running total of penalty time (in seconds) stored on your team's database record. This number increases with every penalty trigger and is never reduced during gameplay.
penaltyMinutes	The admin-controlled session setting that determines how many minutes are added per penalty event. Default is 5 minutes. The admin may change this before or during the event.
Server Penalty	A penalty that is permanently recorded in the database via POST /api/team/add-penalty. This is what counts toward your final score. Not all in-game "errors" trigger a server penalty.
Local Penalty	An in-game penalty that only affects the current puzzle's state (e.g., Game 4's -15 second timer reduction on wrong move). This does NOT add to your server-side penalty total.
Progression Wipe	In Game 2 (Handshake), when repeated wrong actions push the progress bar from above 0 all the way back to 0. A server penalty is only triggered on every 2nd wipe (i.e., count 2, 4, 6...).
hasTriggeredPenalty	A guard flag in Game 1 that prevents the same wrong gate configuration from triggering multiple penalty events. It resets only when the team removes a gate from a slot.
Collision (Game 4)	Occurs in The Firewall when a player attempts a move that violates the protocol rule at their current position, or tries to move outside the 7×7 grid boundary. Costs -15s local timer.
PENDING (Game 7)	The status of the currently active memory request in Game 7. Only the PENDING request can be answered. Correctly answering it unlocks the next LOCKED request.
finalPenalty	The total penalty seconds recorded when the session ends. This is used in the final results to calculate the team's ranking alongside the number of puzzles solved.
Cycle Time	In Game 2 (Handshake), a new hex code is generated every 2,500 ms (2.5 seconds). If the previous code is still pending at cycle end, it auto-fails as a missed code.
Target Score (Game 2)	In Handshake Protocol, the goal is to reach 10 correct signature verifications (progress = 10). Wrong answers remove 2 points; correct answers add 1 point.



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IMPORTANT REMINDERS

Penalties are permanent during the session. Once a server penalty is recorded, it cannot be reversed by the admin or by solving puzzles faster. The best strategy is always to *understand the rules before acting*.

Read the Guide Manual first. Several games (especially Game 2 and Game 6) require knowledge of the protocol rules that are found in the player guide manual. The in-game UI will remind you to consult the manual. Attempting to guess these rules will almost certainly result in penalties.

Each game has 3 sets (except Game 2 which has 2). Penalties apply individually within each set — there is no "carry-over" reset between sets. Your total penalty across all sets is what matters on the leaderboard.

Game 4 is unique — wrong moves only cost local timer seconds, not server penalty. *Only a full timeout* triggers a server penalty in The Firewall. This means you can afford to make move mistakes as long as you complete the path before the timer hits zero.

Game 2 is also unique — the penalty throttle means not every wipe-to-zero triggers a server penalty. Only every *second* complete progress wipe costs a block. Stay consistent and try not to let the bar drain to zero at all.